

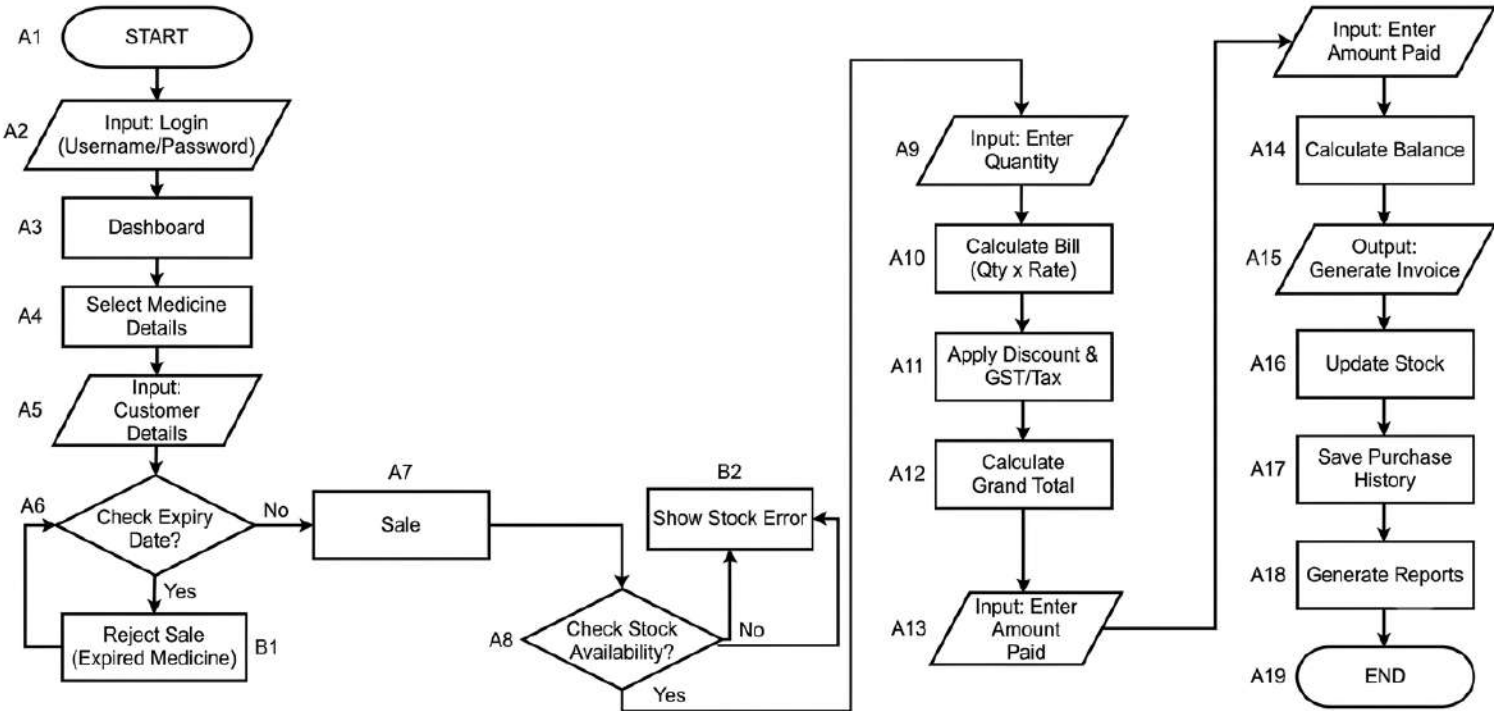
ER Diagram: Pharmacy Management System



Explanation

- Rectangles = Entities, Ovals = Attributes, Diamonds = Relationships, - PK = Primary Key, FK = Foreign Key.
- Relationships and Cardinality: 1:M, 1:1.
- Entity = Pharmacy Medicine, customer_Name, mobile sales, handlers of the variainmation and finacem tradinarid chremic verms.
- Sale_Details = Medicine_Name, company, Medicine_ID (FK), Quantity - Selling_Price, Purchase_Price, Manufacturing_Date, Expiry_Total.
- Medicine = Medicine_Name: acttan,ane, sales and Billing, Amount and Balance, Item_Total.

PHARMACY MEDICINE SALES AND BILLING MANAGEMENT SYSTEM





Pharmacy Medicine Sales and Billing Management System

Step-by-Step Algorithm

Suitable for a college project / practical record.



1 Start and Login



- Start the application.
- Enter username and password.
- Login to the system.

2 Add, update, delete, and search medicine details



- Enter medicine ID, name, company, batch number, stock, price, Mfg. date, Expiry date.
- Allow add, update, delete and search operations.

3 Add, update, delete, and search customer details



- Enter customer ID, name, mobile number, address and purchase history.
- Allow add, update, delete and search operations.

4 Check medicine expiry date and identify expired/near-expiry medicines



- Check expiry dates of medicines.
- Show expired and near-expiry medicines.
- Prevent sale of expired medicines.

5 Prevent the sale of expired medicines



- If medicine is expired, block the sale.
- Show proper alert message.

6 Select customer and medicines for sale



- Choose customer from the list.
- Select medicines and enter quantity.
- Check stock availability.

7 Check available medicine stock



- Verify stock quantity for selected medicines.
- Show alert if stock is insufficient.

8 Enter quantity and calculate item total



- Enter quantity for each medicine.
- Calculate item total using:
Item Total = Quantity × Selling Price

9 Calculate subtotal, discount, GST/tax, grand total, amount paid, and balance



- Calculate subtotal (sum of item totals).
- Apply discount (if any).
- Calculate GST/tax.
- Display grand total.
- Enter amount paid and calculate balance.

10 Generate the bill/invoice



- Show complete bill with all details.
- Print or save the invoice.

11 Automatically reduce medicine stock after a successful sale



- Deduct sold quantity from stock.
- Update the medicine stock.

12 Store the customer's purchase history



- Save each transaction in customer history.
- Allow to view past purchases.

13 Generate daily/monthly sales and revenue reports



- Show daily and monthly sales.
- Calculate total revenue and profit.
- Provide reports with date range.

14 End the process



- Log out from the system.
- Close the application.

